Lesson 3: Variables — Boxes with Names 📦

> **Goal:** Create variables to store values, update them, and use them in calculations.

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1. What is a Variable?

Imagine a **box** with a label on it. You can put a number (or text) inside the box, and use the label to get it back later.

In Python, a **variable** is a named box that stores a value.

> 💡 **Scratch connection:** Variables in Python work exactly like **"Make a Variable"** in Scratch!

Scratch	Python
Make a variable called `score`	`score = 0`
Set `score` to 10	`score = 10`
Change `score` by 1	`score = score + 1`
Say `score`	`print(score)`

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2. Creating a Variable

Use the `=` sign to create a variable and give it a value.

```
age = 8
name = "An"
score = 100
```

- `age` is a variable that stores the number `8`.
- `name` is a variable that stores the text `"An"`.
- `score` is a variable that stores the number `100`.

The `=` sign means **"store this value"**, not "equals" like in math!

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3. Using a Variable

Once you create a variable, you can use it anywhere:

```
age = 8
print(age)          # prints: 8
print("I am", age, "years old")  # prints: I am 8 years old
print(age + 2)      # prints: 10
```

Output:

```
8
I am 8 years old
10
```

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4. Updating a Variable

You can change the value stored in a variable:

```
score = 0
print(score)    # 0

score = 10
print(score)    # 10

score = score + 5   # take the old value, add 5, store back
print(score)    # 15
```

> 💡 `score = score + 5` means: "take the current value of `score`, add 5, and store the result back into `score`."

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5. Variable Naming Rules

Rule	Good ✅	Bad ❌
Use letters, numbers, underscore `_`	`my_score`, `player1`	`my score`, `player-1`
Cannot start with a number	`score1`	`1score`
Case-sensitive	`score` ≠ `Score`	—
No spaces	`high_score`	`high score`

Rule	Good ✅	Bad ❌
Use English words	`total_price`	—

Good variable names:

```
player_name = "An"
total_score = 0
number_of_apples = 5
side_length = 10
```

Bad variable names (will cause errors):

```
1player = "An"      # starts with a number
my score = 0        # has a space
total-score = 0     # has a hyphen
```

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6. Multiple Variables

You can create as many variables as you need:

```
# Rectangle dimensions
width = 15
height = 8

# Calculate area and perimeter
area = width * height
perimeter = 2 * width + 2 * height

print("Width:", width)
print("Height:", height)
print("Area:", area)
print("Perimeter:", perimeter)
```

Output:

```
Width: 15
Height: 8
Area: 120
Perimeter: 46
```

This is much cleaner than writing `print("Area:", 15 * 8)` — if you change the width, you only need to change it in one place!

7. Swapping Two Variables

What if you want to swap the values of two variables?

```
a = 5
b = 10

print("Before:", a, b)   # Before: 5 10

# Swap using a temporary variable
temp = a
a = b
b = temp

print("After:", a, b)    # After: 10 5
```

> 💡 Python also has a shortcut for swapping:
>
> ```
> a, b = b, a
> ```

8. Variable Types

Variables can store different types of values:

Type	Example	Vietnamese
`int`	`age = 8`	Số nguyên
`float`	`height = 1.35`	Số thực
`str`	`name = "An"`	Chuỗi ký tự

```
age = 8            # int
height = 1.35      # float
name = "An"        # str (string)

print(age)         # 8
print(height)      # 1.35
print(name)        # An
```

You can check the type of a variable with `type()`:

```
print(type(8))        # <class 'int'>
print(type(1.35))     # <class 'float'>
print(type("An"))     # <class 'str'>
```

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9. Common Mistakes

```
# Mistake 1: Using a variable before creating it
print(x)        # NameError! x does not exist yet

# Mistake 2: Wrong direction of =
8 = age         # SyntaxError! Left side must be the variable name

# Mistake 3: Forgetting quotes for strings
name = An       # NameError! Python thinks "An" is a variable name
name = "An"     # ✅ Correct
```

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10. Vocabulary

English	Vietnamese	Meaning
variable	biến	A named box that stores a value
assign	gán	Give a value to a variable (`=`)
value	giá trị	The data stored in a variable
update	cập nhật	Change the value of a variable
store	lưu trữ	Save a value in a variable
type	kiểu dữ liệu	The kind of data (int, float, str)
string	chuỗi	Text data, written in quotes
swap	hoán đổi	Exchange the values of two variables
temporary	tạm thời	Used for a short time, then discarded
naming convention	quy tắc đặt tên	Rules for how to name variables

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11. Summary

✅ A variable is a named box that stores a value. ✅ Use `=` to create or update a variable. ✅ Variable names use letters, numbers, and `_`. No spaces. Start with a letter. ✅ You can use a variable in math: `score = score + 10`. ✅ Variables make code easier to read and change. ✅ Types: `int` (whole number), `float` (decimal), `str` (text).

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12. Homework

🟢 Easy

Exercise E1. My Info

Create 3 variables: `name`, `age`, `city`. Assign your own values. Print them.

- **Input:** None.
- **Output:** (example)

```
An
8
Ho Chi Minh City
```

Exercise E2. Rectangle Area

Create two variables `width = 12` and `height = 7`. Calculate and print the area.

- **Input:** None.
- **Output:**

```
84
```

Exercise E3. Update a Variable

Start with `score = 0`. Add 10, then add 5, then subtract 3. Print the final score.

- **Input:** None.
- **Output:**

```
12
```

Exercise E4. Swap

Create `a = 3` and `b = 9`. Swap their values. Print `a` and `b` after the swap.

- **Input:** None.
- **Output:**

```
9
3
```

Exercise E5. Circle Perimeter

The perimeter of a circle is `C = 2 * pi * r`. Use `pi = 3.14159` and `r = 5`. Print the perimeter.

- **Input:** None.
- **Output:**

```
31.4159
```

Exercise E6. Seconds in a Week

Use variables to calculate how many seconds are in one week. Store each step in a variable.

- **Input:** None.
- **Output:**

```
604800
```

(Hint: `seconds_per_minute = 60`, `minutes_per_hour = 60`, `hours_per_day = 24`, `days_per_week = 7`.)

Exercise E7. Type Check

Create one variable of each type: `int`, `float`, `str`. Print each variable and its type.

- **Input:** None.
- **Output:** (example)

```
8
<class 'int'>
```

```
1.75
<class 'float'>
An
<class 'str'>
```

🟡 Medium

Exercise M1. Temperature Converter

Store the temperature in Celsius in a variable `celsius = 100`. Convert it to Fahrenheit using the formula `F = C * 9 / 5 + 32`. Print both values with labels.

- **Input:** None.
- **Output:**

```
Celsius: 100
Fahrenheit: 212.0
```

Exercise M2. Speed, Distance, Time

A car travels at `speed = 60` km/h for `time = 2.5` hours. Calculate the distance and print it.

- **Input:** None.
- **Output:**

```
Distance: 150.0 km
```

Exercise M3. Piggy Bank

You have `savings = 50000` dong. You earn `allowance = 20000` dong per week. After 4 weeks, how much do you have? Print the result.

- **Input:** None.
- **Output:**

```
130000
```

Exercise M4. Average Score

Store 5 test scores in 5 variables: `s1 = 8`, `s2 = 9`, `s3 = 7`, `s4 = 10`, `s5 = 6`. Calculate and print the average.

- **Input:** None.
- **Output:**

```
8.0
```

Exercise M5. Pythagorean Check

Store `a = 3`, `b = 4`, `c = 5`. Calculate `a*a + b*b` and `c*c`. Print both. Are they equal?

- **Input:** None.
- **Output:**

```
a*a + b*b = 25
c*c = 25
```

Exercise M6. Ticket Price

An adult ticket costs `adult_price = 80000` dong. A child ticket costs `child_price = 50000` dong. A family buys `adults = 2` adult tickets and `children = 3` child tickets. Print the total cost.

- **Input:** None.
- **Output:**

```
Total: 310000 dong
```

Exercise M7. Body Mass Index

BMI = weight (kg) / (height (m) × height (m)).

Store `weight = 30` and `height = 1.35`. Calculate and print the BMI (rounded to 2 decimal places is fine).

- **Input:** None.
- **Output:**

```
BMI: 16.46
```

(Hint: `print("BMI:", round(bmi, 2))`)

🔴 Hard

Exercise H1. Buying Fruit

Minh goes to the market. He buys **A** kg of apples at **P1** dong/kg and **B** kg of oranges at **P2** dong/kg. He pays with a **500000-dong** note. How much change does he get?

- **Input:** Four integers on one line: A P1 B P2 (1 ≤ A, B ≤ 10, 1000 ≤ P1, P2 ≤ 50000).
- **Output:** One integer — the change in dong.

Example:

Input	Output
2 15000 3 20000	410000
1 50000 1 50000	400000

Explanation: 2 × 15000 + 3 × 20000 = 30000 + 60000 = 90000. Change = 500000 - 90000 = 410000.

(Hint: Set `a = 2`, `p1 = 15000`, `b = 3`, `p2 = 20000` in your code.)

Exercise H2. Average Speed

A car travels from city A to city B. The first half of the trip (distance **D1** km) is at speed **V1** km/h. The second half (distance **D2** km) is at speed **V2** km/h.

Calculate the **total time** (in hours) for the whole trip.

- **Input:** Four numbers: D1 V1 D2 V2 (1 ≤ D1, D2 ≤ 1000, 1 ≤ V1, V2 ≤ 200).
- **Output:** The total time in hours (as a decimal).

Example:

Input	Output
120 60 80 40	4.0
100 50 100 100	3.0

Explanation: Time = distance / speed. Total time = D1/V1 + D2/V2. For example 1: 120/60 + 80/40 = 2 + 2 = 4.0 hours.

(Hint: `time = d1 / v1 + d2 / v2`.)

Exercise H3. Staircase Tiles

A staircase has **N** steps. Step 1 has 1 tile, step 2 has 2 tiles, ..., step N has N tiles. How many tiles are needed in total?

- **Input:** One integer N (1 ≤ N ≤ 1000).
- **Output:** One integer — the total number of tiles.

Example:

Input	Output
4	10
1	1
10	55

Explanation: For N = 4: 1 + 2 + 3 + 4 = 10. The formula is N × (N + 1) / 2.

(Hint: Use the formula `total = n * (n + 1) // 2`. Set `n = 4` in your code.)

Exercise H4. Savings Goal

Lan wants to save money to buy a bicycle that costs **G** dong. She already has **S** dong saved. She saves **W** dong every week. How many **full weeks** does she need to save the rest?

- **Input:** Three integers: G S W (1 ≤ S < G ≤ 1000000, 1 ≤ W ≤ 100000).
- **Output:** One integer — the number of full weeks needed.

Example:

Input	Output
500000 200000 50000	6
100000 10000 30000	3

Explanation: Lan needs 500000 - 200000 = 300000 more dong. At 50000/week: 300000 / 50000 = 6 weeks.

(Hint: `weeks = (g - s + w - 1) // w` — this rounds up. Or think: `needed = g - s`, then `weeks = needed // w` if it divides evenly, else `weeks = needed // w + 1`.)

Exercise H5. Paint a Wall

A rectangular wall is **W** meters wide and **H** meters tall. One can of paint covers **A** square meters. How many **full cans** of paint are needed to paint the whole wall?

- **Input:** Three integers: W H A (1 ≤ W, H ≤ 100, 1 ≤ A ≤ 500).
- **Output:** One integer — the number of cans needed (round up if not exact).

Example:

Input	Output
10 3 8	4
5 4 10	2
6 6 9	4

Explanation: For 10 × 3 = 30 m². Each can covers 8 m². 30 / 8 = 3.75 → need 4 cans (round up).

(Hint: `cans = (w * h + a - 1) // a` — this is the ceiling division trick.)

Exercise H6. Digit Sum Divisibility

Given a **3-digit** number N, check if the **sum of its digits** is divisible by 3. Print the digit sum and the remainder when divided by 3.

- **Input:** One integer N (100 ≤ N ≤ 999).
- **Output:** Two lines: the digit sum, then the remainder when divided by 3.

Example:

Input	Output
123	6
	0
456	15
	0

Input	Output
124	7
	1

Explanation: For 123: 1 + 2 + 3 = 6. 6 % 3 = 0 (divisible by 3). For 124: 1 + 2 + 4 = 7. 7 % 3 = 1 (not divisible).

(Hint: Extract digits using `//` and `%` from Lesson 2, then add them.)

Exercise H7. Bouncing Ball

A ball is dropped from a height of **H** meters. Each time it bounces, it reaches **half** the previous height. What is the total distance the ball has traveled after **3 bounces** (3 ups + 3 downs + the initial drop)?

- **Input:** One integer H (1 ≤ H ≤ 1000).
- **Output:** The total distance as a decimal number.

Example:

Input	Output
100	275.0
8	22.0

Explanation: For H = 100:

- Drop: 100 m down.
- Bounce 1: 50 m up + 50 m down.
- Bounce 2: 25 m up + 25 m down.
- Bounce 3: 12.5 m up + 12.5 m down.
- Total = 100 + 50 + 50 + 25 + 25 + 12.5 + 12.5 = 275.0 m.

(Hint: Use variables `h1 = H / 2`, `h2 = h1 / 2`, `h3 = h2 / 2`. Total = H + 2*h1 + 2*h2 + 2*h3.)